

Conditional DNA generation by a compact Titans neural-memory model reveals decoder-hidden diversity but incomplete genomic organization

Gonzalo Vidal • SeqTrainer Stage C Study
Exploratory research article; not peer reviewed

Abstract. Neural-memory models update a learned function while processing a sequence, potentially combining short-range attention with an adaptive summary of long-past context. We implemented a paper-traceable Memory-as-Context (MAC) Titans model for bacterial DNA and studied a compact 2.52-million-parameter checkpoint trained on 5.0 million valid bases. The exact two-layer functional memory remained finite, improved a controlled associative objective in all four blocks, and achieved held-out 1.9828 bits per base. Conditional generation initially appeared vocabulary-collapsed under top-k 128. A preregistered nine-policy decoding ablation showed that this collapse was largely decoder-induced: removing top-k increased tokenizer-aligned six-mer diversity from 0.285 to 0.988 of reference and reduced six-mer Jensen–Shannon divergence from 0.545 to 0.326 bits. Nevertheless, unrestricted samples remained 4.70 percentage points GC-poor, had 34% more heuristic ORFs, 57% shorter longest ORFs, 33% lower predicted coding density, 32% shorter median genes and sevenfold longer median intergenic spans. Thus the model learned broad short-range bacterial sequence support and stable online memory dynamics, but not yet realistic gene-scale organization. These exploratory results justify a controlled scale-up, not claims of functional sequence generation or adaptive-memory benefit.

How the tested model reads, remembers and predicts DNA

A compact Memory-as-Context (MAC) network; four identical blocks are stacked.

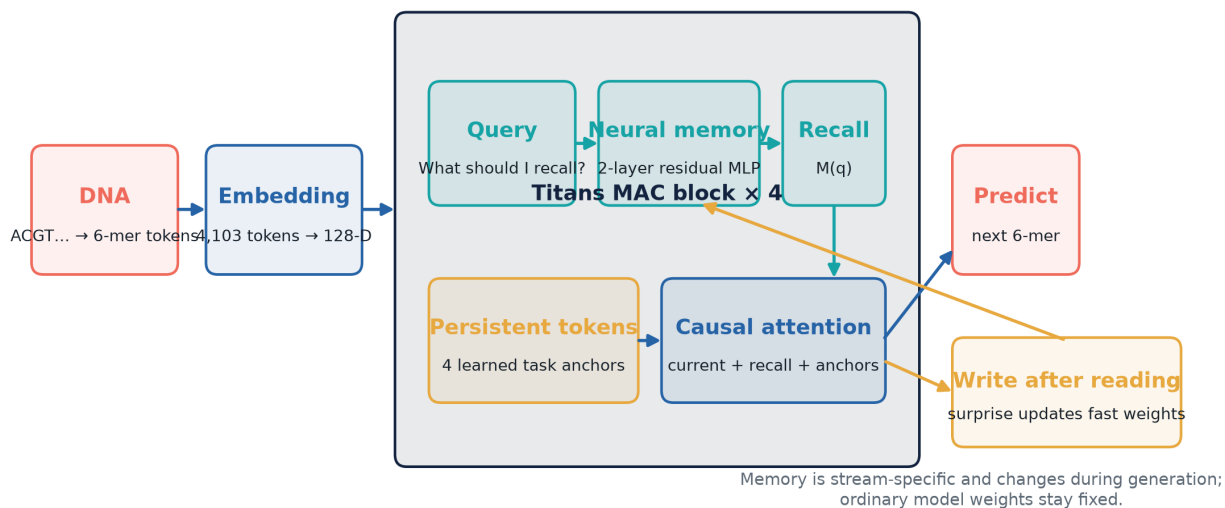


Figure 1 | Accessible view of the tested architecture. Six-base tokens are embedded, passed through four MAC blocks, and decoded autoregressively. Each block retrieves from stream-specific fast weights before attention and writes once after reading the segment.

Introduction

Transformers retain precise short-term context through attention but incur quadratic cost in context length. Behrouz, Zhong and Mirrokni introduced Titans as a family combining a limited-window attention core, a neural long-term memory and persistent learned tokens. The long-term memory is unusual: its own weights are updated online from an associative objective, so the model can compress past context into parameters while it is running [1].

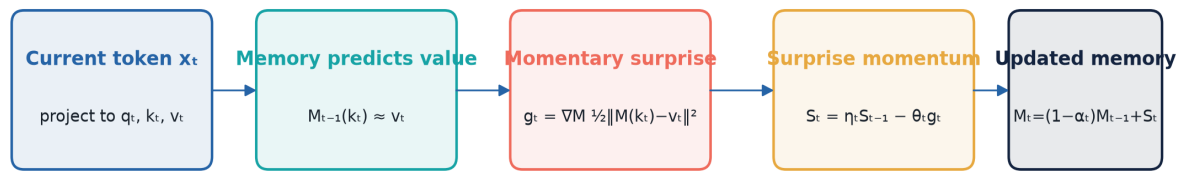
DNA is a stringent test because local composition, codon structure, genes, regulatory intervals and genome-scale organization coexist over different length scales. A generator can look plausible by GC content while failing at higher-order motifs or gene organization. We therefore treat generation as a hierarchy of diagnostics rather than as evidence of biological function.

Study question

Does a compact, paper-deep Titans MAC implementation (i) train and retrieve stably, (ii) retain broad conditional DNA diversity, and (iii) generate sequence statistics resembling held-out *Escherichia coli*? The present work is explicitly exploratory and adaptive-only; it cannot estimate the causal benefit of memory.

Architecture and mathematical correspondence

Neural memory update: prediction error becomes “surprise”



η : how long past surprise persists θ : strength of the new write

α : fraction of old memory forgotten

c16 initializes $\eta=0.9$, $\theta=0.001$ and $\alpha=0.001$, learns them by layer/channel, and applies no surprise cap.

Figure 2 | Associative neural-memory recurrence. Prediction error in the inner memory objective produces momentary surprise; momentum carries recent surprise and adaptive forgetting manages finite capacity.

Paper-defined mechanism

For representation x_t , the Titans paper projects keys, values and queries and trains memory M to associate keys with values:

```
k_t = x_t W_K, v_t = x_t W_V, q_t = x_t W_Q
L_paper(M_{t-1}; x_t) = \sum_j [M_{t-1}(k_t)_j - v_{t,j}]^2
S_t = \eta_t * S_{t-1} - \theta_t * \text{grad}_M L_paper(M_{t-1}; x_t)
M_t = (1 - \alpha_t) * M_{t-1} + S_t, \text{retrieval } y_t = M_{t-1}(q_t)
```

Here $*$ denotes the paper's element-wise, layer/channel-gated multiplication.

Here η controls persistence of past surprise, θ controls the present write, and α controls forgetting. The paper's MAC layout concatenates persistent parameters, the current segment and retrieved history before attention; attention output then writes the long-term memory [1].

SeqTrainer c16 realization

The implementation preserves the same key–value objective and recurrence but makes explicit choices where the paper does not uniquely specify an executable path. Its loss is $\frac{1}{2} \|M(k) - v\|^2$ summed over 128 output channels. The factor $\frac{1}{2}$ leaves the optimum unchanged and removes the factor two from the gradient; learned θ absorbs the scale. Unlike the paper's chunk approximation, c16 computes exact evolving nonlinear gradients token by token with higher-order autograd and truncates the outer graph at horizon three.

Element	Titans paper	c16 implementation
Memory function	MLP, depth ≥ 1	Residual 2-linear-layer MLP: 128→512→128, GELU + LayerNorm
Associative loss	$\sum_j [M(k)_j - v_j]^2$	$0.5 \sum_{j=1}^{128} [M(k)_j - v_j]^2$

Element	Titans paper	c16 implementation
Recurrence	momentum + forgetting	same recurrence; per-layer/output-channel α , η , θ
Initialization	data-dependent gates	$\alpha=.001$, $\eta=.9$, $\theta=.001$; learned thereafter
Projection context	learned Q/K/V	independent Q/K/V + causal depthwise convolution, kernel 4
Normalization	implementation-dependent	L2-normalized q and k
Inner update	chunkwise acceleration described	exact evolving nonlinear gradient; no scan approximation
Numerics	not a biological prescription	FP32 memory; no RMS conditioner; no norm-4 surprise cap

Model tested

Property	c16 value
Parameters	2,518,144 trainable
Blocks / width / heads	4 / 128 / 4
Memory	two-layer residual MLP in each block; expansion 4
Tokenizer	non-overlapping six-mer; vocabulary 4,103
Segment / gradient horizon	32 tokens (192 bases) / 3 segments
Persistent tokens	4 per block
Functional memory state	4,235,264 bytes per active stream
Backend	exact accelerated memory; SDPA; FP32
Training	5,000,060 bases, 8,839 optimizer steps, T4 GPU
Corpus exposure	0.000388 passes of 12.87-billion-base corpus

Methods

Data and training

Training streams were restricted to the Stage C *E. coli* plus *Escherichia* eligibility set, tokenized as non-overlapping six-mers. The model was optimized at learning rate 3×10^{-4} with gradient clipping at 0.5 and weight decay 0.1. The 5M budget is only 0.0388% of one predictable-corpus pass; it is a scale gate, not saturation training.

Held-out and memory analyses

Held-out evaluation covered 38,926 bases in 206 segments from eight streams, but all eight streams belonged to one accession and one ANI99 clade. A controlled 64-pair probe measured immediate reduction in associative reconstruction error and delayed aligned-versus-shuffled recall. Memory-state and hidden-state PCA were exploratory; taxonomic separation could not be evaluated because all plotted streams were *E. coli*.

Conditional generation and decoding ablation

Generation used held-out *E. coli* prompts. The first study generated four 6,144-base continuations per temperature under top-k 128. The follow-up held prompt identity, seed and checkpoint constant while changing one decoding dimension at a time: temperature (0.6–1.1) at k=1,024, top-k (128–unrestricted) at T=0.8, and nucleus p (0.95–1.0) without top-k. Each screen policy used two 3,072-base continuations. Scientific ranking prioritized lower six-mer Jensen–Shannon divergence, then higher aligned-token diversity, then lower GC error.

Metrics

Bits per base (BPB) is cross-entropy normalized by predictable DNA bases; lower is better. Aligned six-mer diversity is the fraction of distinct non-overlapping tokenizer tokens, reported relative to real continuation diversity.

Jensen–Shannon divergence compares k-mer frequency distributions, $JSD(P,Q) = \frac{1}{2}KL(P||R) + \frac{1}{2}KL(Q||R)$, $R = (P+Q)/2$; zero is identical and one bit is maximally disjoint under base-2 logs [3]. Prodigal calls coding sequences computationally; its calls are not functional assays [2]. ORF scans are transparent six-frame start-to-stop heuristics.

Results

Stable learning and active associative memory

Outcome	Observed value	Interpretation
Held-out BPB	1.98278	finite learning signal; narrow one-accession panel
Memory update / surprise	1.4556 / 0.04434	non-zero online writes
Retrieval / state drift	110.76 / 21.78	active finite state; scale-dependent
Gradient interventions	0%	paper-exact recurrence did not invoke a repair
Gate means α / η / θ	.00118 / .89542 / .00124	near intended retention/momentum/write regime
Immediate probe	9.60–11.31% better, all blocks	one write reduces declared associative loss
Delayed probe	positive in 4/4 blocks	pair-specific recall persists after interference

Controlled probes show active associative memory—not biological meaning

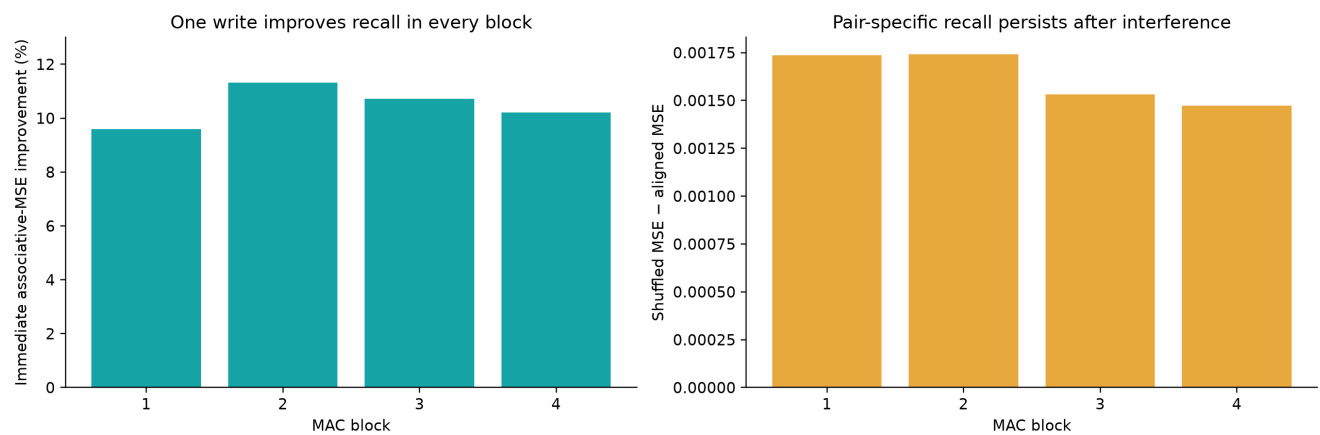


Figure 3 | Controlled memory behavior. Every block improves the inner associative target immediately and retains a small positive aligned-versus-shuffled margin. This validates mechanism, not biological utility.

Decoding, not the model alone, caused the apparent diversity collapse

The apparent generation collapse was largely decoder-induced

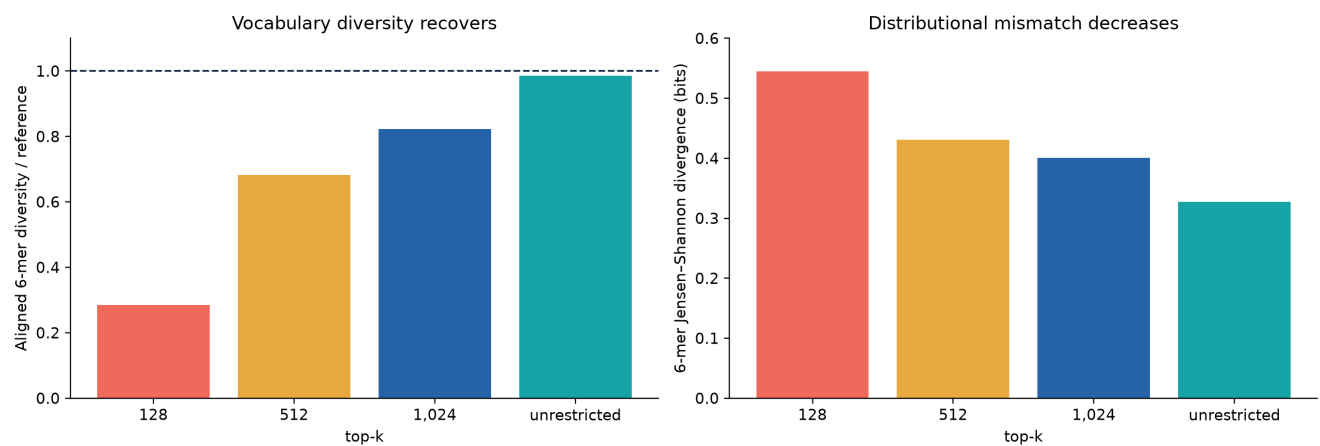


Figure 4 | Top-k ablation. At fixed $T=0.8$ and $p=.99$, removing the hard vocabulary cutoff restores almost all reference-level token diversity and reduces six-mer mismatch by 40%.

top-k	Aligned diversity / ref	3-mer JSD	6-mer JSD	GC absolute error
128	0.285	0.1352	0.5450	0.0745
512	0.682	0.0825	0.4310	0.0594
1,024	0.822	0.0616	0.4010	0.0646
unrestricted	0.986	0.0200	0.3270	0.0474

At k=128, each step retained at most ~3% of the six-mer vocabulary. Unrestricted p=.99 recovered 98.6% of reference aligned diversity. Therefore the 03h collapse cannot be attributed solely to model training. The probability tail beyond rank 1,024 carries substantial, context-dependent sequence diversity.

Temperature and nucleus sampling

At fixed k=1,024, increasing temperature from 0.6 to 1.1 improved diversity only from 0.802 to 0.842 of reference and six-mer JSD from 0.4055 to 0.3933. Temperature 0.6 was therefore not beneficial. With no top-k, p=.95, .99 and 1.0 yielded diversity ratios .982, .986 and .988 and six-mer JSD .3335, .3270 and .3261. Full p=1.0 sampling is the least biased scientific diagnostic; p=.99 is a nearly equivalent practical policy.

Unrestricted samples remain genomically miscalibrated

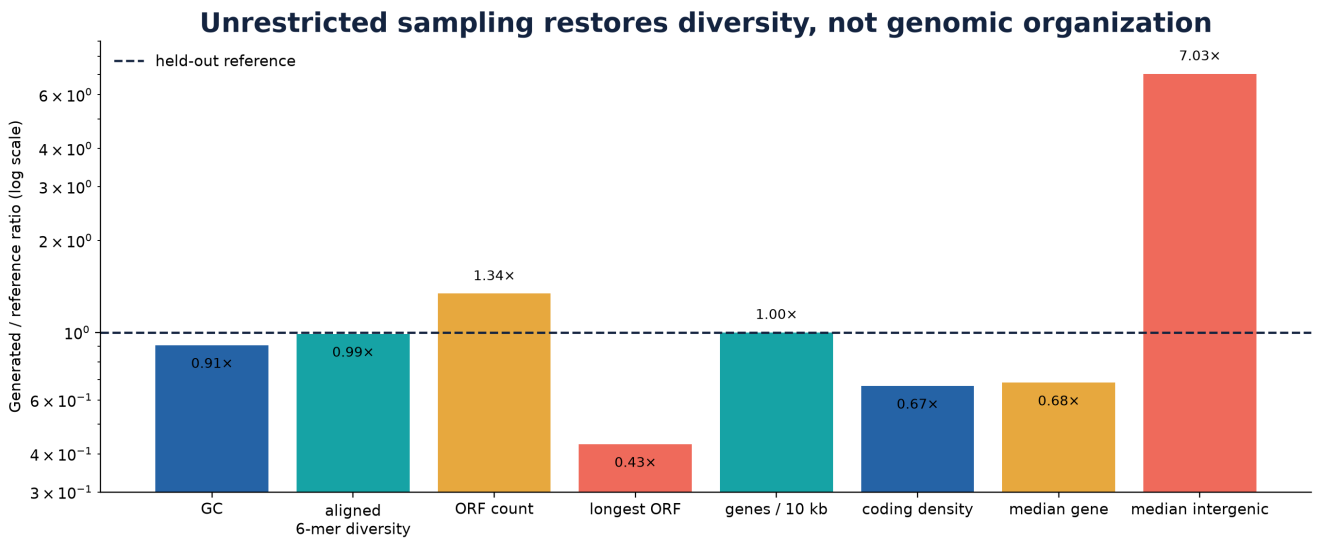


Figure 5 | Biological calibration of the best decoding policy. Ratios are generated/reference and use a logarithmic axis. Token diversity and gene-call rate match; gene architecture does not.

Metric	Reference	T=.8, no-k, p=1	Meaning
GC fraction	0.5142	0.4671	4.70 percentage points AT-biased
Aligned diversity	0.9219	0.9111	0.988× reference; no vocabulary collapse
Overlapping diversity	0.6384	0.6572	1.029×; diversity ≠ correctness
ORFs ≥90 bp	28.0	37.5	1.34×; excess short coding-like spans
Longest ORF	864 bp	372 bp	0.43×; poor long coding coherence
Genes / 10 kb	8.14	8.14	same count, only five calls/group
Coding density	0.494	0.330	0.67×
Median gene	504 bp	345 bp	0.68×
Median intergenic	72 bp	506 bp	7.03×

Short k-mer distributions were close (JSD=0.0016, JSD=0.0080, JSD=0.0192), but mismatch rose with motif length (JSD=0.3261). A reference-versus-reference finite-sample baseline is absent, so the absolute excess over natural variation is unknown. Prodigal gene counts matched, but predicted genes were shorter and more widely separated. The generator learned coding-like local patterns, not convincing gene-scale organization.

Discussion

What the findings demonstrate

The study demonstrates a technically functional deep neural memory: exact nonlinear surprise gradients remain finite, online writes are non-zero, all blocks improve a controlled associative target, and broad decoding modestly increases surprise without destabilizing the state. It also demonstrates that the c16 static model distribution contains broad six-mer support and comparatively accurate one- to four-mer statistics. These are meaningful engineering and exploratory learning results.

What they do not demonstrate

Evidence ladder: where the present study stops

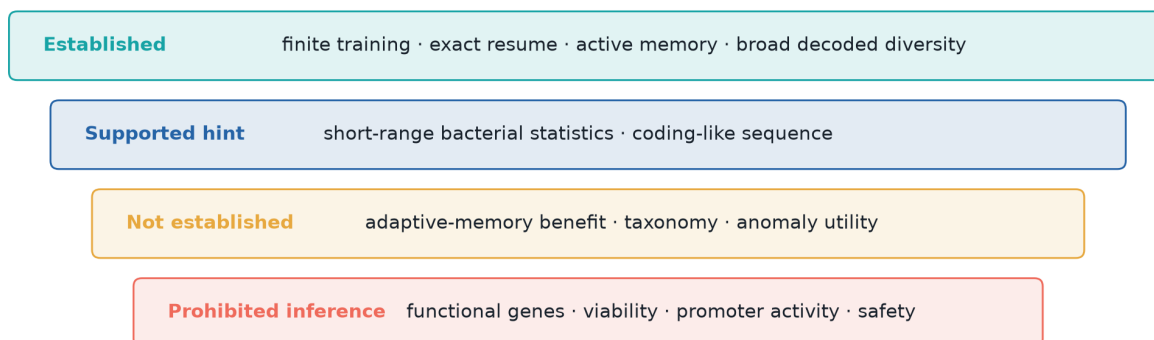


Figure 6 | Claim boundary. The current evidence ends at stable mechanism and conditional sequence diagnostics.

No matched no-memory checkpoint was trained at 5M, so neither BPB nor generation can be causally attributed to adaptive memory. Prodigal calls and ORFs do not establish expression, protein function, promoters, operons, viability, fitness or safety. PCA from one accession cannot establish taxonomic structure. Two-prompt decoding screens do not support confidence intervals or generalization claims. The generated fragments are conditional continuations, not de novo chromosomes.

Advantages and liabilities

Advantages observed	Liabilities observed
Expressive two-layer memory with exact, stable updates	Higher-order gradients and per-stream fast state are computationally costly
Memory remains active over long generation	Benefit over static/no-memory architecture is unmeasured
Unrestricted distribution is broad, not collapsed	Hard top-k can severely and misleadingly suppress diversity
Short-range bacterial statistics are learned	GC and gene/intergenic organization remain miscalibrated
Append-only protocol and artifact hashing	Validation panel is narrow; ledger recorder has a commit-context defect

Future work and decision path

First, confirm $T=.8$, no top-k, $p=1.0$ with four accessions and 1,024 tokens, adding $p=.99$ as a practical secondary policy. Compute split-half reference JSD, GC-matched random and six-mer Markov baselines, codon usage, frame periodicity, stop-codon frequency and full gene/intergenic distributions. Second, freeze a clade/accession-stratified held-out panel. Third, qualify the 24.8M-parameter Medium model (12 blocks, $d=256$, eight heads) on A100 with FP32 functional memory and an empirically selected batch. Fourth, train one adaptive-only 25M discovery run with gates at 5M, 10M and 25M. Extend to 50M only if diverse-panel BPB, GC, k-mer JSD and gene organization improve while memory remains finite. Finally, only after the adaptive model is useful should matched no-memory, frozen-memory, shallow-memory and second-seed controls be run; only those controls can support a memory-benefit claim.

Data and code availability

The checkpoint SHA-256 is 21898362291f4fd1e6aafcfcbe47e8b05dbe69e5c8036e6ae7927a6ac24ac4541. Drive-backed ledger events record the 5M run, first generation study and decoding ablation. Generation code was pinned to commits fc36e9f287240b3ad3367a092a7983f9376bff5b and 6d08cd8a3128d4d61a7e67ab20eebaa0dbff48b5. The article package is generated from source under docs/stage_c_c16_article.

References

- [1] Behrouz A, Zhong P, Mirrokni V. Titans: Learning to Memorize at Test Time. Advances in Neural Information Processing Systems 39 (2025). https://proceedings.neurips.cc/paper_files/paper/2025/file/a4ca07aa108036f80cbb5b82285fd4b1-Paper-Conference.pdf
- [2] Hyatt D et al. Prodigal: prokaryotic gene recognition and translation initiation site identification. BMC Bioinformatics 11, 119 (2010). doi:10.1186/1471-2105-11-119.
- [3] Lin J. Divergence measures based on the Shannon entropy. IEEE Transactions on Information Theory 37, 145–151 (1991).
- [4] Vaswani A et al. Attention Is All You Need. Advances in Neural Information Processing Systems 30 (2017).
- [5] SeqTrainer Stage C paper-deep implementation decision and frozen protocol, repository evidence dated 26–31 July 2026.

Reporting note

This manuscript follows an evidence-tiered reporting policy. Exact engineering, held-out prediction, controlled associative behavior, representation exploration and generation diagnostics are separated. No p-values are reported because the decoding screen contains only two paired prompts and was not designed for inferential testing.